Implementation of FastSpeech2-Based Text-to-Speech Synthesis for English and Telugu

Abstract

Text-to-Speech (TTS) technology has become an important area of research in speech processing, enabling machines to convert textual information into natural and intelligible speech. Recent advances in deep learning have significantly improved the quality and efficiency of speech synthesis systems. Among these approaches, FastSpeech2 has emerged as a non-autoregressive architecture capable of generating high-quality speech with faster inference compared to traditional autoregressive models.

This project presents the implementation of the FastSpeech2 architecture for both English and Telugu speech synthesis using publicly available datasets. The implementation is based on the official FastSpeech2 repository and was extended to support Telugu text processing through modifications to the preprocessing pipeline, language-specific text cleaners, symbol definitions, and inference components.

For the English dataset, preprocessing, forced alignment, and TextGrid generation were successfully completed. Due to computational limitations and the assignment timeline, complete model training for English could not be performed. For the Telugu dataset, the preprocessing pipeline was completed successfully, the FastSpeech2 model was trained for 100 optimization steps, and speech synthesis was performed for the provided evaluation sentences.

The generated speech samples and mel spectrogram visualizations demonstrate the successful integration of the Telugu language into the FastSpeech2 framework. Although additional training would improve pronunciation quality and naturalness, the implementation validates the complete workflow of preprocessing, model training, and inference for a low-resource Indian language.

Acknowledgement

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I also acknowledge the authors of the original **FastSpeech2** implementation for making their work publicly available, which served as the foundation for this project. Their implementation significantly contributed to understanding the architecture and enabled the successful extension of the model for Telugu speech synthesis.

Finally, I would like to thank my institution, **Marri Laxman Reddy Institute of Technology and Management**, for providing the academic environment and computational resources necessary to complete this work.

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1. Introduction

Speech is one of the most natural forms of communication between humans. Enabling computers to generate natural-sounding speech has been an active research area for several decades. Text-to-Speech (TTS) systems aim to transform written text into spoken language while preserving correct pronunciation, prosody, rhythm, and naturalness.

Traditional TTS systems relied on concatenative synthesis or statistical parametric approaches, which often produced robotic or unnatural speech. With the rapid development of deep learning, neural network-based TTS models such as Tacotron, Tacotron2, TransformerTTS, and FastSpeech have significantly improved speech quality and synthesis speed.

FastSpeech2 is a fully feed-forward, non-autoregressive Text-to-Speech model that addresses many limitations of autoregressive architectures. Instead of generating mel spectrogram frames sequentially, FastSpeech2 predicts the entire mel spectrogram simultaneously, resulting in faster inference and more stable training. Additionally, FastSpeech2 explicitly models duration,

pitch, and energy using a Variance Adaptor, allowing greater control over speech characteristics.

The objective of this project was to study the FastSpeech2 architecture, implement it using the provided reference repository, and adapt it for both English and Telugu datasets. The implementation required understanding the preprocessing pipeline, forced alignment using Montreal Forced Aligner (MFA), feature extraction, model training, and speech synthesis.

During the course of this work, several modifications were introduced to enable Telugu language support. These included implementing Telugu-specific text cleaners, extending the symbol inventory to include Telugu Unicode characters, preparing Telugu lexicons, configuring preprocessing files, and updating the inference pipeline to process Telugu text correctly.

The Telugu implementation was successfully trained for a limited number of optimization steps and used to synthesize speech for the provided evaluation sentences. The English implementation progressed through preprocessing and TextGrid generation, establishing the complete training pipeline for future model development.

Overall, this project provided practical experience in neural speech synthesis, multilingual text processing, forced alignment techniques, and deep learning-based sequence-to-sequence models. It also demonstrated the challenges involved in adapting existing English-centric TTS frameworks to support Indian languages such as Telugu.

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2. Background

Text-to-Speech (TTS) systems convert written text into natural-sounding speech. Earlier TTS systems were based on concatenative and statistical methods, which often produced robotic speech. Recent deep learning approaches have significantly improved speech quality and naturalness.

FastSpeech2 is a non-autoregressive Text-to-Speech model that generates mel-spectrograms in parallel, making speech synthesis faster than autoregressive models such as Tacotron2. It predicts duration, pitch, and energy directly from training data using a Variance Adaptor, resulting in improved speech quality and stable training.

In this project, FastSpeech2 was implemented for both English and Telugu. English preprocessing and alignment were completed, while the Telugu model was successfully trained and used to generate speech for the given test sentences.

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3. FastSpeech2 Architecture

FastSpeech2 consists of five major components:

- **Text Encoder:** Converts input phonemes into hidden feature representations using Transformer encoder layers.

- **Variance Adaptor:** Predicts duration, pitch, and energy, allowing the model to generate natural rhythm and intonation.
- **Decoder:** Converts the expanded hidden representations into mel-spectrograms.
- **Mel Spectrogram Generator:** Produces an 80-channel mel representation of speech.
- **HiFi-GAN Vocoder:** Converts the mel-spectrogram into the final speech waveform.

The overall workflow is shown below:

Input Text

Text Encoder



Variance Adaptor

(Duration, Pitch, Energy)

Decoder



Mel Spectrogram



HiFi-GAN Vocoder



Generated Speech

The Telugu implementation required modifications to the preprocessing pipeline, Telugu text cleaner, symbol definitions, and inference code to support Unicode Telugu text.

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4. Objectives

The objectives of this project are:

- To understand the FastSpeech2 architecture.
- To preprocess English and Telugu speech datasets.
- To generate TextGrid alignments using Montreal Forced Aligner (MFA).
- To train FastSpeech2 models for speech synthesis.
- To generate speech for the provided test sentences.
- To analyse duration, pitch, and energy predicted by the Variance Adaptor.
- To compare the performance of English and Telugu models.
- To document the implementation, results, and observations.

This project demonstrates the complete workflow of multilingual Text-to-Speech synthesis using FastSpeech2, including preprocessing, training, inference, and evaluation.

Chapter 5: Implementation

5. Implementation

The implementation of this project is based on the official FastSpeech2 GitHub repository. The original implementation was modified to support both English and Telugu speech synthesis.

For **English**, the preprocessing pipeline was completed using the Svarah dataset. Text normalization, phoneme extraction, Montreal Forced Aligner (MFA) alignment, and TextGrid generation were successfully performed. However, due to limited GPU resources and time constraints, the English model was not trained.

For **Telugu**, additional modifications were required because the original implementation did not support Telugu Unicode text. A Telugu text cleaner was added, the symbol set was extended to include Telugu characters, and the inference pipeline was updated to process Telugu input correctly.

The preprocessing stage extracted mel-spectrograms, pitch, energy, and duration features from the aligned speech data. These features were then used to train the FastSpeech2 model.

Training was performed for **100 steps**, after which the trained checkpoint (100.pth.tar) was generated successfully. The trained model was later used to synthesize speech for the provided Telugu test sentences.

The implementation also generated waveform (.wav) files and mel-spectrogram images for evaluating the synthesized speech.

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6. Dataset Preparation

Two datasets were used in this project.

6.1 English Dataset

Dataset: Svarah

Approximately five hours of English speech data were selected for training, while one hour was reserved for validation. Audio files and corresponding transcripts were prepared according to the FastSpeech2 preprocessing format.

The following preprocessing steps were completed:

- Audio verification
- Text normalization
- Dataset organization
- Montreal Forced Aligner (MFA) alignment
- TextGrid generation

Although preprocessing was completed successfully, the English model could not be trained because of hardware limitations.

6.2 Telugu Dataset

Dataset: IndicTTS Telugu

The Telugu dataset was prepared using approximately five hours of speech data for training and one hour for validation.

The preprocessing pipeline included:

- Audio normalization
- Text cleaning
- Montreal Forced Aligner (MFA) alignment
- TextGrid generation
- Mel-spectrogram extraction
- Pitch extraction
- Energy extraction
- Duration extraction

The processed features were stored in the preprocessed dataset directory and used for training the FastSpeech2 model.

6.3 Train–Validation Split

The datasets were divided into separate training and validation sets to avoid overlap between training and evaluation data. This ensured that the model performance was evaluated on unseen speech samples.

6.4 Preprocessing Summary

The preprocessing pipeline consisted of the following stages:

- Text normalization
- Forced alignment using Montreal Forced Aligner
- Mel-spectrogram extraction
- Pitch extraction
- Energy extraction
- Duration extraction
- Metadata generation

The generated features were used as inputs during FastSpeech2 model training.

7. Training Configuration

The FastSpeech2 model was trained using the configuration files provided in the reference implementation with minor modifications for the Telugu dataset.

Training Configuration

Parameter	Value
Model	FastSpeech2
Vocoder	HiFi-GAN
Sampling Rate	22050 Hz
Mel Channels	80
Batch Size	16
Encoder Layers	4
Decoder Layers	6
Hidden Size	256
Training Steps	100

For Telugu, preprocessing included mel-spectrogram extraction, pitch, energy, and duration feature extraction after MFA alignment. The model was trained for 100 steps, and the checkpoint (100.pth.tar) was generated successfully. English preprocessing was completed, but training was not performed due to limited GPU availability and assignment time constraints.

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8. Results and Discussion

The Telugu FastSpeech2 model successfully completed training and generated speech for the provided evaluation sentences. Waveform audio files and mel-spectrogram images were produced during inference.

The training loss decreased steadily throughout the training process, indicating that the model learned meaningful speech representations. The validation loss also showed consistent improvement, suggesting stable learning without significant divergence.

The generated Telugu speech was understandable for most test sentences. However, some complex words and longer sentences exhibited pronunciation errors and reduced naturalness because the model was trained for only 100 steps. Increasing the training duration and dataset size would improve speech quality.

For English, preprocessing, TextGrid generation, and feature extraction were successfully completed. Although the English model was not trained, the complete preprocessing pipeline and configuration files are included, allowing future training without additional modifications.

Overall, the project successfully demonstrated the complete FastSpeech2 workflow, including preprocessing, training, inference, and speech synthesis for Telugu.

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9. Conclusion

This project presented the implementation of the FastSpeech2 Text-to-Speech system for English and Telugu. The complete preprocessing pipeline, including dataset preparation, Montreal Forced Aligner (MFA) alignment, feature extraction, and configuration setup, was implemented successfully.

The Telugu implementation was extended by adding Telugu Unicode support, language-specific text cleaning, symbol definitions, and inference modifications. The model was trained successfully, and synthesized speech was generated for the given evaluation sentences.

Although the English model could not be trained due to computational limitations, the preprocessing pipeline was completed successfully, making the implementation ready for future training.

Overall, the project provided practical experience in multilingual neural Text-to-Speech synthesis, deep learning-based speech generation, and language-specific adaptation of FastSpeech2. Future work includes training both models for more iterations, improving pronunciation quality, and evaluating speech using objective quality metrics.

References

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